

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [150, 149, 154, 172, 156, 183, 138, 170, 197, 131]

Tap amplitudes (pixels): [122.41323457861898, 118.81077392223317, 110.43550153822818, 114.85643212288984, 107.07474025184464, 107.07474025184464, 104.04326023342406, 116.21101496846157, 109.85900054160332, 136.12494260788506, 98.71676655968832]

GPT-5

MDS-UPDRS Item 3.4 Finger Tapping – Scoring Analysis

Given data

- Reported number of taps: 11 (i.e., one extra tap beyond the instructed 10)
- Tap periods (frames) for intervals between taps 1 → 2 through 10 → 11: [150, 149, 154, 172, 156, 183, 138, 170, 197, 131]
- Tap amplitudes (pixels) for taps 1 → 11: [122.41, 118.81, 110.44, 114.86, 107.07, 107.07, 104.04, 116.21, 109.86, 136.12, 98.72]

Note: The standard instruction is to rate on 10 taps. Since 11 amplitudes are present, I will:

- Use the first 10 taps (amplitudes 1–10) to assess amplitude trend and interruptions across the first 10 taps (consistent with instructions).
- Use the 10 inter-tap periods provided to assess rhythm/hesitations within that 10-tap window.

Rhythm/Interruptions (hesitations/halts)

- Inter-tap periods vary: 131–197 frames. Largest prolongation at 197 and 183 suggests hesitations.

- Counting “interruptions” as conspicuous prolongations relative to the local pace, there appear to be multiple rhythm breaks:
 - Notable longer intervals: 172, 183, 170, 197 (four clear hesitations vs surrounding intervals).
- This aligns with “3 to 5 interruptions” → consistent with score 2 (Mild). It may border higher if the 197-frame interval reflects a longer arrest, but based on provided data alone it looks like repeated hesitations rather than a freezing halt.

Speed (slowing)

- Periods generally trend upward mid-sequence (e.g., 154 → 172 → 156 → 183), indicating slowing and/or hesitations.
- This is more than “slight” but not clearly “moderate” continuous slowing; pattern is inconsistent with persistent marked bradykinesia but shows mild-to-moderate variability.

Amplitude and decrement

Using the first 10 taps (per instruction):

- Amplitudes 1–10: 122.41, 118.81, 110.44, 114.86, 107.07, 107.07, 104.04, 116.21, 109.86, 136.12
- Early taps show a downward drift from tap 1 to 3, slight recovery at 4, further decrease to 7, recovery at 8–10 with a large increase by tap 10.
- This is not a monotonic decrement starting after the 1st tap (which would indicate score 3 for decrement criterion). Nor is it a clear decrement beginning midway and persisting; instead it’s variable with recovery. Therefore, amplitude criterion alone would not push to 2 or 3; if anything, it argues against a higher score.

Overall interpretation mapped to criteria

- Interruptions: approximately 3–5 notable hesitations → aligns with score 2 (Mild).
- Speed: mild slowing/variability → compatible with 1–2; combined with interruptions, supports 2.
- Amplitude: variable with recovery; does not support 3’s “decrements starting after the 1st tap,” nor 2’s “decrements midway” strongly, but amplitude doesn’t lower the score given interruptions.

Final score